1 Model

Since helium recombination occurs much deeper the envelope, we restrict our analysis to regions where hydrogen is the dominant species. Assuming LTE, the total number density of particles is $n = \frac{\rho}{m_p}(1+x)$, where the ionization fraction $x \equiv n(H^+)/n(H)$ is given by the Saha equation:

$$\frac{n(e^-)n(H^+)}{n(H^0)} = \frac{x^2}{1-x}\frac{\rho}{m_p} = \frac{(2\pi m_e kT)^{3/2}}{h^3}e^{-\epsilon_i/kT}. \tag{1}$$

We assume that hydrogen recombination is adiabatic, with the released energy contributing directly to the internal energy of the gas. In this case, the total pressure and internal energy are

$$p = (1+x)\frac{\rho}{m_p}kT + \frac{aT^4}{3}, \tag{2}$$

$$u = (1+x)\frac{3}{2}\frac{kT}{m_p} + \frac{aT^4}{\rho} + \frac{\epsilon_i x}{m_p}. \tag{3}$$

The momentum equation is

$$v\frac{dv}{dr} + \frac{1}{\rho}\frac{dp}{dr} + \frac{GM_r}{r^2} = 0, \tag{4}$$

where $\frac{dM_r}{dr} = 4\pi r^2\rho$, and the pressure gradient is given by

$$\frac{1}{\rho}\frac{dp}{dr} = f_1\frac{d\ln T}{dr} + f_2\frac{d\ln\rho}{dr}, \tag{5}$$

$$f_1 = \frac{T}{\rho}\left.\frac{\partial p}{\partial T}\right|_\rho = \frac{(1+x)kT}{m_p}\left[1 + 4\alpha + \frac{x(1-x)}{(2-x)(1+x)}\left(\frac{3}{2} + \frac{\epsilon_i}{kT}\right)\right], \tag{6}$$

$$f_2 = \left.\frac{\partial p}{\partial\rho}\right|_T = \frac{(1+x)kT}{m_p}\left[1 - \frac{x(1-x)}{(2-x)(1+x)}\right], \tag{7}$$

$$\alpha = \frac{p_{\rm rad}}{p_{\rm gas}}. \tag{8}$$

From mass conservation ($\dot{M} = 4\pi r^2\rho v$), we have

$$\frac{d\ln\rho}{dr} = -\frac{2}{r} - \frac{d\ln v}{dr}. \tag{9}$$

Combining Eqs. (4), (5) and (9) gives

$$\boxed{(v^2 - f_2)\frac{d\ln\rho}{d\ln r} = f_1\frac{d\ln T}{d\ln r} + \frac{GM_r}{r} - 2v^2}. \tag{10}$$

Radiation diffusion introduces a cooling term:

$$du = \frac{p}{\rho^2}d\rho - \frac{1}{\rho}\nabla\cdot\vec{F}_{\rm rad}\,dt = \frac{p}{\rho^2}d\rho - \frac{1}{\rho}\left[\frac{1}{r^2}\frac{d}{dr}(r^2F_{\rm rad})\right]\frac{dr}{v} = \frac{p}{\rho^2}d\rho - \frac{1}{\dot{M}}dL_{\rm rad}, \tag{11}$$

where the radiative flux is

$$F_{\rm rad} = \frac{L_{\rm rad}}{4\pi r^2} = -\frac{4acT^3}{3\kappa_{\rm rad}\rho}\frac{dT}{dr} \qquad \rightarrow \qquad \boxed{\frac{d\ln T}{d\ln r} = -\frac{3\kappa_{\rm rad}\rho}{4acT^4}\frac{L_{\rm rad}}{4\pi r}} \tag{12}$$

Combining Eqs (2), (3) and (11) gives

$$\boxed{\frac{1}{\dot{M}}\frac{dL_{\rm rad}}{d\ln r} = f_1\frac{d\ln\rho}{d\ln r} - f_3\frac{d\ln T}{d\ln r}}, \tag{13}$$

$$f_3 = \left.\frac{\partial u}{\partial\ln T}\right|_\rho = \frac{(1+x)kT}{m_p}\left[\frac{3}{2} + 12\alpha + \frac{x(1-x)}{(2-x)(1+x)}\left(\frac{3}{2} + \frac{\epsilon_i}{kT}\right)^2\right]. \tag{14}$$

2 Critical-point condition

The critical-point condition depends on how efficiently radiation diffuses relative to the outflow. From Eq. (13), we define

$$Q \equiv \frac{d\ln T}{d\ln r} - \frac{f_1}{f_3}\frac{d\ln \rho}{d\ln r} = -\frac{1}{\dot{M} f_3}\frac{dL_{\rm rad}}{d\ln r}, \tag{15}$$

which measures the departure from adiabaticity. In the strictly adiabatic limit, $dL_{\rm rad}/dr = 0$ and hence $Q = 0$. In this case, T is slaved to ρ according to

$$\frac{d\ln T}{d\ln r} = \frac{f_1}{f_3}\frac{d\ln \rho}{d\ln r}. \tag{16}$$

We can rewrite Eq. (10) as

$$\left(v^2 - f_2 - \frac{f_1^2}{f_3}\right)\frac{d\ln \rho}{dr} = \frac{GM_r}{r^2} - \frac{2v^2}{r}. \tag{17}$$

The critical point condition is therefore

$$v^2 = c_{s,\rm ad}^2 = f_2 + \frac{f_1^2}{f_3}, \qquad \frac{GM_r}{r} - 2\left(f_2 + \frac{f_1^2}{f_3}\right) = 0. \tag{18}$$

In the opposite limit, when radiative diffusion is rapid compared with advection, the temperature is no longer determined primarily by the local adiabatic response to the density. Instead, its gradient is set by diffusion in Eq. (12). In this case, $d\ln T/dr$ acts as an independently determined source term in Eq. (10), and the critical point occurs when

$$v^2 = f_2, \qquad -f_1\frac{3\kappa_{\rm rad}\rho}{4acT^4}\frac{L_{\rm rad}}{4\pi r} + \frac{GM_r}{r} - 2f_2 = 0. \tag{19}$$

3 Three regimes

3.1 Zone I: quasi-adiabatic (QA)

When $Q \to 0$, Eq. (13) becomes numerically stiff because it is evaluating a small quantity as a difference of two nearly identical large numbers. Since Q is small, we can calculate it algebraically instead. To zeroth order, $Q_0 = 0$ and

$$L_0 = -\frac{16\pi acrT^4}{3\kappa_{\rm rad}\rho}\frac{f_1}{f_3}\left(\frac{d\ln \rho}{d\ln r}\right)_0, \qquad \left(\frac{d\ln \rho}{d\ln r}\right)_0 = \frac{GM_r/r - 2v^2}{v^2 - (f_2 + f_1^2/f_3)}. \tag{20}$$

According to Eq. (15), the first-order correction of diffusion is

$$Q \simeq -\frac{1}{\dot{M} f_3}\frac{dL_0}{d\ln r} + \mathcal{O}(\epsilon^2), \tag{21}$$

$$\frac{d\ln \rho}{d\ln r} \simeq \frac{f_1 Q + GM_r/r - 2v^2}{v^2 - (f_2 + f_1^2/f_3)}, \tag{22}$$

$$\frac{d\ln T}{d\ln r} \simeq \frac{f_1}{f_3}\frac{d\ln \rho}{d\ln r} + Q. \tag{23}$$

where $\epsilon \sim L_{\rm rad}/\dot{M}v^2$. The treatment above fails when the near-adiabatic Q-expansion is no longer justified, i.e., when $\eta = (L - L_0)/L_0 \sim 1$.

3.2 Zone II: diffusive (FLD)

Let's consider a small perturbation in $L_{\rm rad}$. Holding other variables constant, we have

$$\frac{d(\delta L)}{d\ln r} = \dot{M}\left[f_1\delta\left(\frac{d\ln\rho}{d\ln r}\right) - f_3\delta\left(\frac{d\ln T}{d\ln r}\right)\right] = \frac{3\dot{M}\kappa_{\rm rad}\rho}{16\pi acT^4 r}\left(f_3 - \frac{f_1^2}{v^2 - f_2}\right)\delta L_{\rm rad}. \tag{24}$$

Therefore, a perturbation in $L_{\rm rad}$ will grow as $\exp\left(\int \lambda_L dr\right)$, where

$$\lambda_L = \frac{3\dot{M}\kappa_{\rm rad}\rho f_3}{16\pi acT^4 r}\left(\frac{v^2 - c_{s,\rm ad}^2}{v^2 - f_2}\right). \tag{25}$$

Near the diffusive critical point, $\lambda_L \to \infty$, and its value is generally much larger than λ_ρ and λ_M. As a result, any tiny numerical error in L (e.g., from the quasi-adiabatic treatment) will grow quickly unbound, making outward integration hopeless while the solution itself is perfectly smooth.

As a result, we stop the quasi-adiabatic treatment when

$$\eta(r_{\rm match}) = 0.01, \qquad \eta \equiv (L - L_0)/L_0. \tag{26}$$

At this radius $r_{\rm match}$, T and L contain small errors of order η^2. The quasi-adiabatic approximation has reached the edge of the accuracy we want, while the system is still being consistent with having negligible diffusion. We then impose a BVP, where we require

$$\rho_{\rm QA} = \rho_{\rm FLD}, \quad T_{\rm QA} = T_{\rm FLD}, \quad L_{\rm FLD} + \dot{M}\mathcal{B}_{\rm FLD} = \dot{E} \qquad \text{at } r = r_{\rm match}, \tag{27}$$

where $\mathcal{B} = v^2/2 + h + \Phi$, and $\dot{E}$ is conserved. In specific,

$$\dot{E} = \dot{M}\mathcal{B}_{\rm base} + L_{\rm core}, \tag{28}$$

where we assume at the inner base the star supplies an outward radiative luminosity $L_{\rm core}$.

Meanwhile, the variables L and T are allowed to change together while preserving the total energy. At the outer boundary, we impose the photosphere condition,

$$\tau = \frac{2}{3}, \quad q \equiv \frac{L}{4\pi acr^2T^4} = \frac{1}{4}, \quad \frac{dL}{dr} = 0 \qquad \text{at } r = r_{\rm ph}, \tag{29}$$

while leaving the location of the photosphere $r_{\rm ph}$ to be free. Therefore, in the BVP, the solver simultaneously determines $\{T_{\rm match}, L_{\rm match}, r_{\rm ph}\}$.

3.3 Zone III: free-streaming

This regime is not yet implemented in the model.

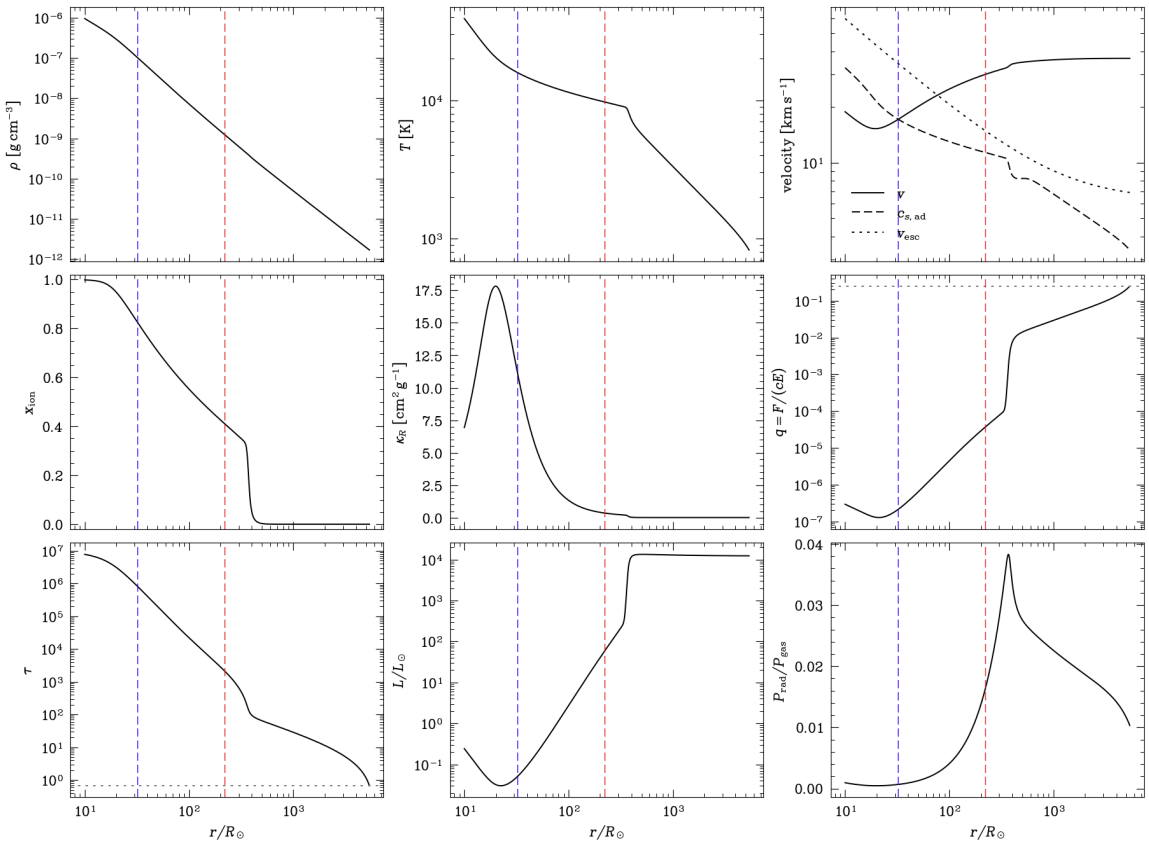


Figure 1: The initial conditions are $\rho_c = 10^{-7}\,\mathrm{g\,cm^{-3}}$, $T_c = 10^{4.2}\,\mathrm{K}$, $M_c = 0.1\,\mathrm{M}_\odot$, and $L_{\mathrm{core}} = 1\,\mathrm{L}_\odot$.

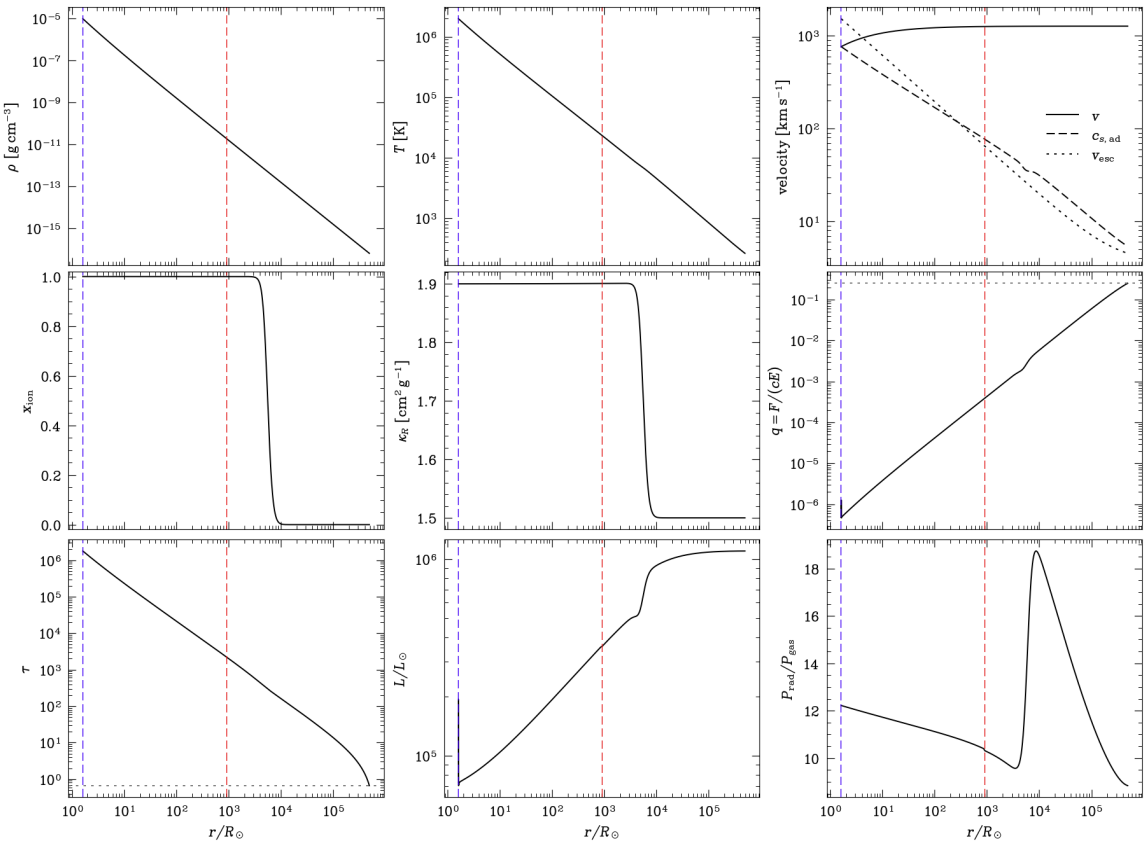


Figure 2: The initial conditions are $\rho_c = 10^{-5}\,\mathrm{g\,cm^{-3}}$, $T_c = 2 \times 10^6\,\mathrm{K}$, $M_c = 10\,\mathrm{M}_\odot$, and $L_{\mathrm{core}} = 7.3 \times 10^4\,\mathrm{L}_\odot$.